

Clinical Research Assistant — Executive Feature Guide

One AI platform spanning the entire clinical-research lifecycle — from the first literature search to regulatory-grade trial execution to the published manuscript and patient results letter.

Audience: research directors · sponsors · institutional leadership · procurement • **Companion:** hands-on *walkthrough guides* (for researchers & investigators) demonstrate each tool step-by-step.

The problem it solves

A single trial moves through **six phases over months or years** — evidence synthesis, trial design, regulatory start-up, execution, analysis & reporting, dissemination. Historically each phase needs its own software (reference managers, EDC, randomisation, statistics, submission tooling, medical writing). Teams lose more time **re-typing the same study into each silo** than doing the science, and **every handoff between tools is an audit and compliance risk**.

What CRA is

One platform organised around *how research actually flows*, not around what each piece of software does. Two complementary halves sit on a **single data spine**:

- 🔍 **Research & Evidence** — find, appraise, synthesise, and publish evidence.
- 📋 **Trial Management & Execution** — design, run, capture, and report a regulated trial.

The same PICO that scopes a meta-analysis **seeds** the statistical analysis plan. The same subject randomised in the eCRF flows automatically into the CDISC submission, the Clinical Study Report, and the patient results letter — **with no re-typing and a continuous audit trail**.

The differentiator in one line: most tools *manage* the workflow; CRA *does the methodological work* — composing search queries, running meta-analyses in a sandboxed compute environment, deriving CDISC datasets, drafting ICH-E3 reports — while enforcing anti-fabrication guardrails at every step.

Lifecycle coverage at a glance



Phase	What leadership gets	Headline deliverables
01 Evidence synthesis	A defensible evidence base for protocols, grants, HTA & guideline submissions	Search strategy · SR protocol · screening · meta-analysis · GRADE/PRISMA
02 Trial design	A powered, peer-review-ready trial design	Sample-size calculation · ICH-E9 Statistical Analysis Plan
03 Start-up	Faster regulatory & ethics approval	Trial-registration drafts (CT.gov / EU CTIS) · IRB packet + consent form
04 Execution	Regulatory-grade data capture at scale	eCRF/EDC · ePRO · randomisation · safety · drug accountability · lab feeds
05 Analysis & reporting	Submission- and publication-ready outputs	CDISC SDTM→ADaM→TLF · CSR · manuscript · lay summary
06 Infrastructure	Governance, oversight & cost control	RBAC · audit · portfolio & budget dashboards · multi-tenant deployment



Research & Evidence — highlights

Capability	Benefit to the institution
Multi-source search strategy	Reproducible, registration-ready Boolean queries across PubMed, Europe PMC, Embase, Cochrane, Scopus, Web of Science in one pass.
Systematic-review screening	Blinded dual review + AI-assist + auto PRISMA flow — collapses the single biggest SR time-sink.
Meta-analysis (pairwise, network, IPD)	Pooled effects, heterogeneity, funnel/Egger, and gold-standard individual-patient analyses — all run in a sandbox, never invented.
GRADE + PRISMA 2020	Journal- and guideline-grade certainty ratings and reporting checklists, computed not asserted.
Living-review watches	Scheduled re-search with materiality alerts and committee-quorum voting — keeps guidelines current automatically.
Manuscript & lay-summary drafters	IMRaD manuscripts with reviewer-response loops, and reading-level-graded patient summaries.



Trial Management & Execution — highlights

Capability	Benefit to the institution
AI-assisted eCRF design	CRFs drafted from a protocol in minutes, with CDASH naming, edit-checks, and immutable versioned publishing.
EDC capture + queries	Hard checks block bad data; soft checks auto-raise queries; every change is audited (old→new, who, when).
Randomisation / IRT	Blinded allocation, multiple algorithms, PI-only emergency unblinding — fully auditable.
Safety (AE/SAE)	ICH-E2A auto-classification, 24-hour reporting timer, FDA 3500A drafting.
Operations suite	Recruitment/CONSORT funnels · visit scheduling & reminders · drug accountability · EHR & lab-feed ingestion.
Multi-site coordination	Per-site KPI rollups for enrolment, query backlog, safety, and operations across an entire programme.

Compliance & governance — built in, not bolted on

Standard	What CRA enforces
21 CFR Part 11 / ALCOA+	Append-only audit trails, e-signature with password re-authentication, immutable published forms.
ICH E6 (GCP)	Role-based access, delegation logs with PI countersignature, training-record tracking, separation-of-duties at user setup.
ICH E9 / E9(R1)	SAPs structured around the estimands framework; statistics run in a controlled compute environment.
ICH E3 · E2A	Standardised Clinical Study Reports and adverse-event safety workflows.
CDISC (SDTM / ADaM / Define-XML / ODM)	Submission-ready datasets and form definitions exported in regulator-expected formats.
21 CFR §50.25 · PRISMA · GRADE · CONSORT	Consent-form element checks, and reporting standards baked into evidence outputs.

Data governance: PHI (clinical) and research data live in **separate stores**; the platform is **multi-tenant** (many institutions, isolated), with a **formal IQ/OQ/PQ validation pack** available for inspection readiness.

Why CRA is different

- **Single data spine** — one study, entered once, flows across all six phases. No re-typing, no reconciliation.
- **Anti-fabrication by design** — the system refuses to invent effect sizes, PMIDs, statistics, or registry IDs; numbers trace to a source or a sandboxed computation.
- **Audit-by-default** — every clinical write is captured; reports are regulator-formatted.
- **Cost transparency** — per-study and per-organisation usage and spend dashboards.
- **One platform, not a suite** — replaces a stack of disconnected tools and the integration burden between them.

40+

integrated capabilities

6

lifecycle phases, one platform

1,500+

automated tests

15+

regulatory standards enforced

In short

The Clinical Research Assistant lets an institution run the **complete research lifecycle on one governed platform** — accelerating evidence synthesis and trial start-up, hardening data capture and safety to inspection standard, and producing submission- and publication-ready outputs — while a continuous audit trail and built-in compliance reduce both cycle time and regulatory risk. Researchers get a tireless co-investigator; leadership gets oversight, defensibility, and a lower total cost of ownership.